

6.4 电磁波极化

电磁波极化是指电磁波在传播过程中电场矢量方向随时间的变化。

1) 极化分类

电磁波极化有	线极化	垂直线极化, 水平线极化
	圆极化	左旋圆极化, 右旋圆极化
	椭圆极化	左旋椭圆极化, 右旋椭圆极化

2) 均匀平面波的两个垂直分量

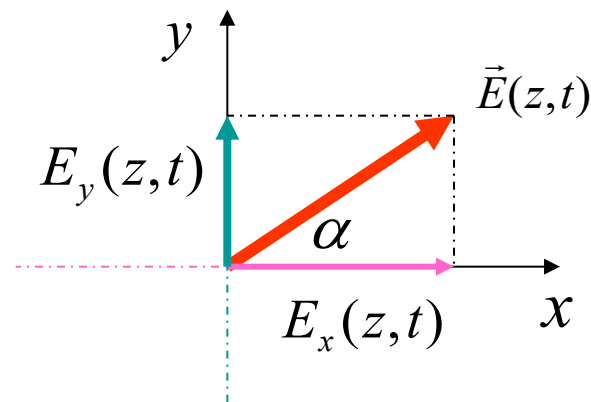
对于沿 z 方向传播的均匀平面波 $\vec{E}(z, t) = \hat{x}E_x(z, t) + \hat{y}E_y(z, t)$

$$E_x(t) = \sqrt{2}E_{x0} \cos(\omega t - kz + \varphi_x)$$

$$E_y(t) = \sqrt{2}E_{y0} \cos(\omega t - kz + \varphi_y)$$

$$E_x = E_{x0} e^{j\varphi_x} e^{-jkz}$$

$$E_y = E_{y0} e^{j\varphi_y} e^{-jkz}$$



根据两个垂直分量振幅和相位的关系, 分以下几种情况

3) $\varphi_y = \varphi_x \pm n\pi$ 两分量同相或反相

$$E_x(t) = \sqrt{2}E_{x0} \cos(\omega t - kz + \varphi_x)$$

$$E_y(t) = \sqrt{2}E_{y0} \cos(\omega t - kz + \varphi_x \pm n\pi)$$

$$= \pm \sqrt{2}E_{y0} \cos(\omega t - kz + \varphi_x)$$

$$E_x = E_{x0} e^{j\varphi_x} e^{-jkz} \quad n \text{ 为偶数取正, 为奇数取负}$$

$$E_y = \pm E_{y0} e^{j\varphi_x} e^{-jkz}$$

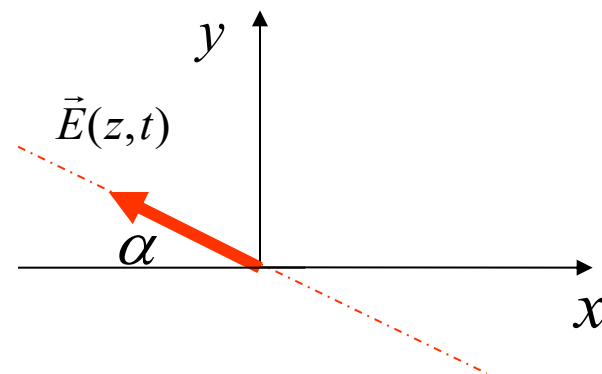
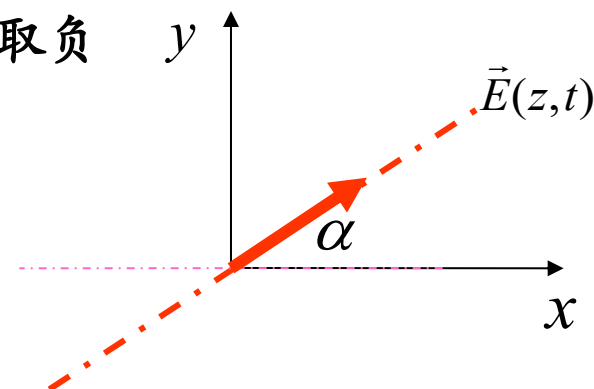
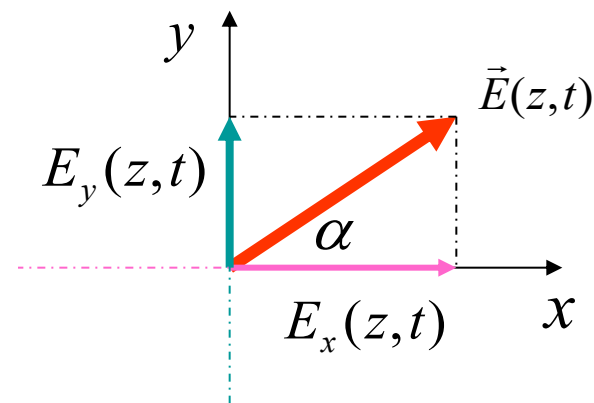
$$\tan \alpha = \frac{E_y(z, t)}{E_x(z, t)} = \pm \frac{E_{y0}}{E_{x0}}$$

$$\vec{E}(z, t) = \sqrt{2}(\hat{x}E_{x0} \pm \hat{y}E_{y0}) \cos(\omega t - kz + \varphi_x)$$

$$|\vec{E}(z, t)| = \sqrt{2} \sqrt{E_{x0}^2 + E_{y0}^2} \cos(\omega t - kz + \varphi_x)$$

线极化 $\varphi_y = \varphi_x \pm n\pi$ n 为偶数取正

n 为奇数取负



$$4) \quad E_{x0} = E_{y0} = E_0 \quad \text{且} \quad \varphi_y = \varphi_x \pm \frac{\pi}{2}$$

$$E_x(t) = \sqrt{2}E_0 \cos(\omega t - kz + \varphi_x)$$

$$E_y(t) = \sqrt{2}E_0 \cos(\omega t - kz + \varphi_x \pm \frac{\pi}{2})$$

$$= \mp \sqrt{2}E_0 \sin(\omega t - kz + \varphi_x)$$

$$E_x = E_0 e^{j\varphi_x} e^{-jkz}$$

$$E_y = E_0 e^{j\varphi_x} e^{\pm j\pi/2} e^{-jkz} = \pm jE_x$$

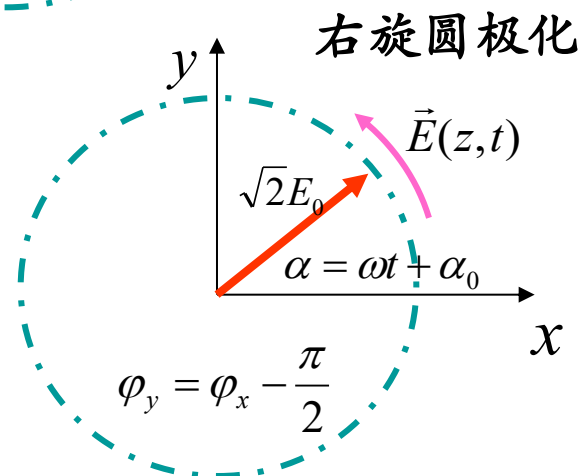
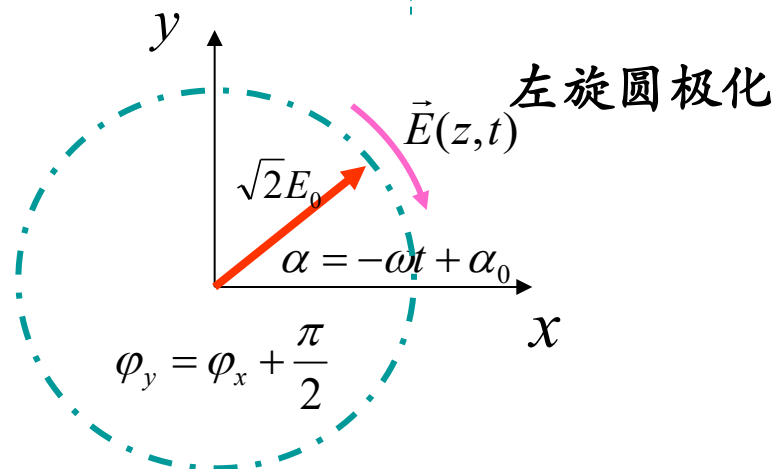
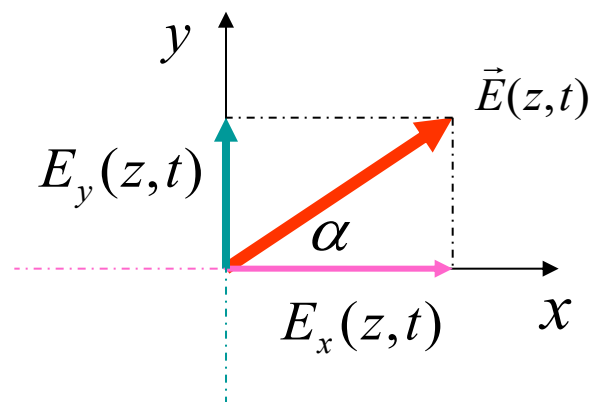
$$\tan \alpha = \frac{E_y(z,t)}{E_x(z,t)} = \mp \tan(\omega t - kz + \varphi_x)$$

$$\alpha = \mp(\omega t - kz + \varphi_x) + n\pi$$

$$|\vec{E}(z,t)| = \sqrt{E_x^2(z,t) + E_y^2(z,t)} = \sqrt{2}E_0$$

$$\varphi_y = \varphi_x + \frac{\pi}{2} \quad \alpha = -\omega t + \alpha_0 \quad \text{左旋圆极化}$$

$$\varphi_y = \varphi_x - \frac{\pi}{2} \quad \alpha = \omega t + \alpha_0 \quad \text{右旋圆极化}$$

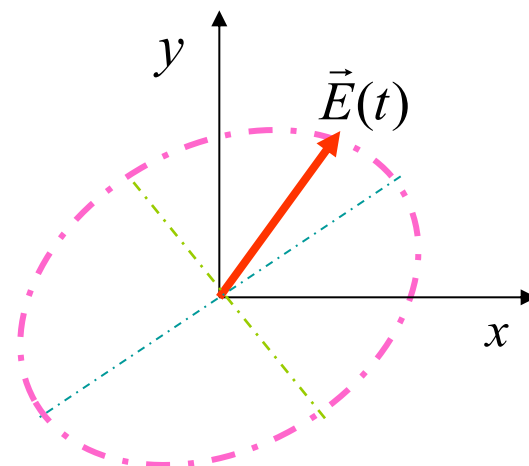


5) 椭圆极化

如果均匀平面波的两个垂直分量之间不符合线极化和圆极化的关系时是椭圆极化。

椭圆极化也有左旋椭圆极化和右旋椭圆极化

线极化和圆极化是特殊的椭圆极化

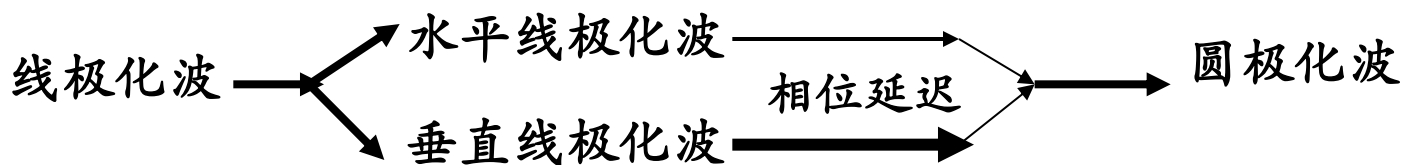
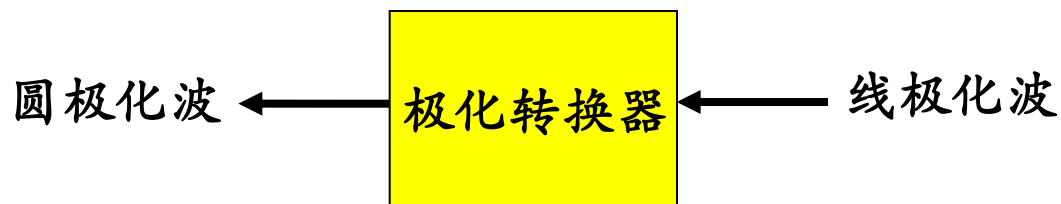


6) 极化的分解与合成





$$\vec{E} = \hat{x}E_0e^{-jkz} = \underbrace{\left\{ \left(\hat{x}\frac{1}{2}E_0 - \hat{y}\frac{j}{2}E_0 \right) \right\}}_{\text{右旋圆极化波}} + \underbrace{\left\{ \left(\hat{y}\frac{j}{2}E_0 + \hat{x}\frac{1}{2}E_0 \right) \right\}}_{\text{左旋圆极化波}} e^{-jkz}$$



水平线极化天线仅能接收水平线极化波

左旋圆极化天线仅能接收左旋圆极化波

圆极化波应用:卫星通信,雷达(克服雨衰,某线极化不敏感反射等效效应)